Feedback Design of an Underactuated Three-link bipedal robot

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Abstract

This paper presents the modeling and control of a three-link bipedal robot (torso and two legs) for stable locomotion. The full nonlinear dynamics are derived using the Lagrangian method, capturing inertia, Coriolis, and gravity terms. A hybrid model accounts for the impulsive ground contact at heel strike. To generate a stable walking gait, we employ Hybrid Zero Dynamics (HZD) with virtual constraints defined by Bézier polynomials, linking actuated joint angles to a cyclic variable. A nonlinear optimization problem is then solved to find Bézier coefficients and initial conditions that yield a periodic orbit. Gait efficiency is evaluated via cost of transport, and the optimized solution ensures the system returns to the same state after each step, producing stable and energetically consistent locomotion. [1]

1 Introduction

Bipedal locomotion is one of the most fundamental challenges in robotics due to various challenging aspects to it such as it's non-linear, underactuated and hybrid nature. Unlike most conventional locomotion methods, robots that walk must continuously maintain balance, while periodically transitioning between continuous motion and discrete events like the impact the foot makes on the ground. These make the analysis and control of bipedal systems more complex.

The common approach to modeling walking robots is to simplify them to a multi link, rigid body system, in the sagittal plane. In this report, a three link bipedal model made up of a torso and two legs is assumed. The system goes through alternating phases of motion; a continuous swing phase, during which one leg moves forward while the other leg remains in contact with the ground, and an impact phase, where the swing leg strikes the ground and becomes the new stance leg. The switching nature between continuous and discrete dynamics makes this a hybrid system.

Designing a stable walking gait for this system requires accurate dynamic modeling, a control system that can handle underactuation, and impacts. Traditional trajectory tracking will fail to provide stability across steps, especially if there are disturbances or non-nonlinearities. To address this, the Hybrid Zero Dynamics (HZD) has proven to be a powerful tool for gait design and stabilization. Instead of directly controlling the system's states, we use HZD to enforce virtual constraints that determine joint motion, thereby reducing the system to a lower dimensional manifold where stable periodic behavior can be worked with. [1]

In our project, we have parameterized virtual constraints using Bézier polynomials, allowing for smooth and flexible trajectory generation. The reduced order dynamics produced can be used to look into the internal behavior of the system and check for the existence of stable periodic orbits. We also use an optimization based approach to find suitable parameters that give us an efficient and repeatable walking gait.

2 Dynamics Model

The three link bipedal model used in this project is modeled as a planar system consisting of a torso and two legs as shown in Fig. 1.

The model is made with the assumptions: (i) All

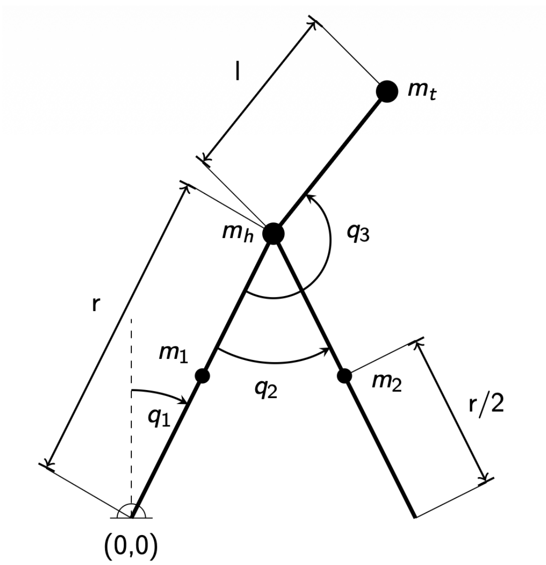

Figure 1: Three-link planar biped model consisting of a torso and two legs.

links are rigid bodies with concentrated masses, (ii) The motion is limited to the sagittal plane, (iii) The stance foot maintains a fixed, no slip contact with the ground during the swing phase, and (iv) impacts between the swing foot and the ground are instantaneous and inelastic.

2.1 Coordinate Definitions

The configuration of the system is described by the generalized coordinates

$$q = [q_1, q_2, q_3]^T, \tag{1}$$

where q_1(cyclic coordinate) represents the absolute angle of the stance leg with respect to the vertical axis of the fixed coordinate frame pivoted at the end of the stance leg, q_2 is the relative angle of the swing leg with respect to the stance leg, and q_3 is the relative angle of the torso with respect to the stance leg.

All angles are defined such that counterclockwise rotation is positive.

The corresponding generalized velocities are given by $\dot{q} = [\dot{q}_1, \dot{q}_2, \dot{q}_3]^T$.

2.2 Equations of Motion

The dynamics of the system are derived using the Lagrangian formulation. The total kinetic and potential energies are used to construct the Lagrangian, and the Euler–Lagrange equations are applied to find the equations of motion.

The resulting dynamics can be expressed in the form

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Bu, \tag{2}$$

where $D(q)$ is the inertia matrix, $C(q, \dot{q})$ represents the Coriolis and centrifugal effects, and $G(q)$ is the gravity vector. The matrix B maps the control inputs u to the generalized coordinates.

Due to passive degrees of freedom, the system is underactuated, with actuation applied at the hip joints.

2.3 State-Space Representation

Defining the state vector as

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \tag{3}$$

the system can be written in state-space form as

$$\dot{x} = \begin{bmatrix} \dot{q} \\ D^{-1}(q)\left(-C(q, \dot{q})\dot{q} - G(q) + Bu\right) \end{bmatrix}. \tag{4}$$

3 Gait Design

The gait design of the underactuated legged system can be simplified by breaking down the full dynamics into a zero dynamics subspace, one that clearly defines the progression of the gait in terms of a cyclic variable s. During the swing phase, the stance leg remains fixed at the ground contact point, acting as a pivot. The system evolves continuously under the equations of motion, with the swing leg moving forward to prepare for the next ground contact. The stance leg angle q_1 serves as a monotonically evolving phase variable that parametrizes the entire step, progressing from an initial value $q_{1,0}$ at the beginning of the step to a final value $q_{1,f}$ just before impact. This makes q_1 a natural choice for the cyclic coordinate. Starting from the full Euler–Lagrange equations of motion, the system is partitioned into actuated and un-actuated joints. Since q_1 is passive (no torque acts directly on the stance leg), it serves as the un-actuated coordinate. The remaining coordinates q_2 and q_3 are actuated at the hip joints. The equations of motion can be written in partitioned form as

$$\begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_a \end{bmatrix} + \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_a \end{bmatrix} + \begin{bmatrix} G_1 \\ G_a \end{bmatrix} = \begin{bmatrix} 0 \\ u \end{bmatrix}, \tag{5}$$

where $q_a = [q_2, q_3]^T$ are the actuated coordinates. When the virtual constraints are satisfied and $y = 0$, the control input u is chosen to enforce these constraints, and the full dynamics reduce to the behavior of the un-actuated coordinate alone.

3.1 The Zero Dynamics Manifold

Enforcing the virtual constraints $y = h(q) = 0$ and $\dot{y} = 0$ restricts the system to an invariant surface called the zero dynamics manifold $\mathcal{Z}$. On this manifold, q_a and $\dot{q}_a$ are entirely determined by q_1 and $\dot{q}_1$ through the constraint relations. The system thus reduces to a scalar second-order ODE in q_1:

$$\ddot{q}_1 = \psi(q_1, \dot{q}_1), \tag{6}$$

which can equivalently be expressed using the variable $z = \dot{q}_1$ as

$$z \frac{dz}{dq_1} = \psi(q_1, z). \tag{7}$$

This one-dimensional representation is the zero dynamics of the system and fully captures the internal behavior of the walking gait.

3.2 Lie Derivative Formulation

To enforce $\dot{y} = 0$ and solve for the feedback linearizing control input, the time derivative of the output is expanded using Lie derivatives. The first time derivative of the output is

$$\dot{y} = L_f h(q) + L_g h(q)\, u, \tag{8}$$

where $L_f h$ and $L_g h$ are the Lie derivatives of h along the drift vector field f and the input vector field g, respectively. When $L_g h(q)$ is invertible (i.e., the decoupling matrix is full rank), the control law

$$u = (L_g h)^{-1} (-L_f h + v) \tag{9}$$

renders $\ddot{y} = v$, where v is an auxiliary PD input driving $y \to 0$. This feedback linearization enforces the virtual constraints and confines the trajectory to $\mathcal{Z}$.

3.3 Impact Model and Gait Reset

At the end of each swing phase, the swing foot strikes the ground, triggering a discrete transition. The configuration q is assumed continuous through impact, but the velocities change instantaneously. Using conservation of angular momentum and the rigid impact assumption, the post-impact velocities are related to the pre-impact velocities by

$$\dot{q}^+ = \Delta(q^-)\, \dot{q}^-, \tag{10}$$

where $\Delta(q^-)$ is the impact map derived from the constrained impact equations. Following the velocity update, a relabeling map R is applied to swap the stance and swing leg identities:

$$(q^+, \dot{q}^+) = (Rq^-, R\,\Delta(q^-)\dot{q}^-), \tag{11}$$

which restores consistency in the coordinate definitions for the next step. Within the zero dynamics manifold, the impact map reduces to a scalar reset of the form

$$z^+ = \delta(q_1^-)\, z^-, \tag{12}$$

where δ is the scalar reduction of Δ projected onto $\mathcal{Z}$. A periodic gait exists when the composition of the continuous zero dynamics flow and this impact reset returns the system to its initial state, closing the hybrid limit cycle.

3.4 Optimization

To compute a reasonable gait, optimization was used to determine optimal values for the zero dynamics and the Bezier coefficients for both q_2 and q_3. The list of optimized parameters has the following form: The optimization vector is defined as:

$$f_0 = [q_{1,0},\ \dot{q}_{1,0},\ \alpha_3,\ \alpha_4,\ \alpha_5,\ \gamma_3,\ \gamma_4,\ \gamma_5] \tag{13}$$

where:

- $q_{1,0}$: pre-impact initial angle of q_1
- $\dot{q}_{1,0}$: pre-impact initial angular velocity of q_1
- $\alpha_{3,5,}$: 3rd to 5th Bezier coefficients for q_2
- $\gamma_{3,5}$: 3rd to 5th Bezier coefficients for q_3

As can be observed in Eq. 13, the following eight parameters need to be selected for optimization. The values chosen for the initial guess were based on the desired outcome of the robot while walking.

Using this logic, $q_{1,0}$, which represents the angle between the swing leg (pre-impact) and the world frame, was selected to be close to 19° (0.325 radians). This value was chosen under the premise that the angle should not be too large, which could cause the robot to exhibit a falling motion, nor too small, which would result in an unrealistic gait. Since these values are later optimized, the selected ϵ_{q_1} for the upper and lower bounds was chosen to be relatively small (0.1 rad), ensuring that the optimized gait remains reasonably close to the initial guess.

For $\dot{q}_{1,0}$, the value was chosen based on the assumption that the gait angular velocity must be consistent with q_1. Therefore, both values are selected to be negative at the beginning of the gait. The chosen velocity was selected to control how fast or slow the gait evolves, ensuring consistency throughout the trajectory. The selected $\epsilon_{\dot{q}_1}$ was chosen to be larger, as the gait velocity significantly affects the robot's motion. Allowing a wider range of possible values (1.0 rad/s) helps the optimizer find a more suitable solution.

Alpha α values were selected following that the bezier coefficients that describe how q_2 evolves during the entire step and measures the angle between the stance leg and the swing leg. During the step the swing leg starts behind the stance leg and moves forward (decreasing q_2, and getting close to it until it passes it and increasing q_2 taking it back to its original value. Based on that fact it was found that at both the beginning and end of the step q_2 was at its highest possible value and towards the middle it was at its lowest value, therefore creating a V-shape of possible values to select for $\alpha_{3,5}$, selecting α_3 as the smallest.

For γ, the values were selected based on the fact that the Bezier coefficients describe how q_3 evolves throughout the entire step and represent the angle between the stance leg and the torso. During the step, the torso starts inclined forward, making the angle smallest at the beginning/end of the step (due to the cyclical nature of the gait). The angle q_3 then increases as the step progresses until it reaches its midpoint, after which it decreases again toward the end of the step. This creates a $\wedge$-shaped profile for the possible values of γ_3 to γ_5, with γ_5 selected as the smallest value.

3.5 Gait Efficiency

[Insert mechanical cost of transport (MCOT) calculations and results.] @Prathyush

$$J_1(\alpha) := \frac{1}{p_2^h(q_0)} \int_0^{T_I(\xi_2)} \|u_\alpha^*(t)\|_2^2 \, dt \tag{14}$$

where $p_2^h(q_0)$ is the horizontal displacement of the swing foot at the moment of impact, and $u(t) \in \mathbb{R}^m$ is the vector of joint torques computed from the Bézier-parameterized virtual constraints. Normalizing by step length allows meaningful comparison across gaits of differing stride lengths, penalizing solutions that expend disproportionate control effort per unit of forward progress. Since the cost penalizes the squared ℓ^2-norm of the control action at each instant, the optimizer is encouraged to distribute torque smoothly across the gait evolution rather than relying on impulsive, intermittent corrections, promoting energetically consistent locomotion. Gaits with poor Bézier parameterization, for instance, those requiring large corrective torques near heel-strike to satisfy the periodicity constraints, incur a disproportionately high energy cost due to the quadratic penalty structure, naturally driving the optimizer away from mechanically inefficient limit cycles. The values given by the optimizer are:

$$f = \begin{bmatrix} -0.64896 & -2.2698 & 0.10266 & 0.21016 \\ 0.72138 & 3.0566 & 3.0742 & 3.0025 \end{bmatrix} \tag{15}$$

3.6 Target Trajectory Generation

The controller for the three-link biped model requires the controller to track a target trajectory and to minimize the error between the actual and target cycle variable's value.
The robot model is

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Bu, \text{ where:} q \in \mathbb{R}^N \tag{16}$$

is the configuration vector, and u is the actuator input.

3.6.1 Virtual Constraint Formulation

The output is defined as,

$$y = h(q) = h_0(q) - h_d(\theta(q)) \tag{17}$$

where: $h_0(q)$ is the actual joint motion, $h_d(\theta)$ is the desired trajectory, $\theta(q)$ is a monotonic phase variable. The goal of a controller is to drive $y \to 0$, that forces the robot to follow the desired gait. Partitioning the H matrix as,

$$h_0(q) = H_0 q$$

and

$$\theta(q) = cq$$

where H_0 is a constant matrix that selects the $(N-1)$ coordinates to be regulated, and c is a constant row vector that defines a scalar quantity evolving monotonically during the step. With this choice, the virtual constraint output becomes

$$y = h(q) = H_0 q - h_d(\theta(q)).$$

This structure is attractive because it depends only on the robot configuration q, making the controller independent of explicit time and naturally synchronized with the motion of the robot. Here,

$$H = \begin{bmatrix} H_0 \\ q \end{bmatrix}$$

3.6.2 Trajectory Generation with Bezier Polynomials

For $1 \leq i \leq N-1$, a one-dimensional Bézier polynomial of degree M is a function

$$b_i : [0,1] \to \mathbb{R},$$

defined by $M+1$ coefficients α_k^i, $k = 0, 1, \ldots, M$, as

$$b_i(s) = \sum_{k=0}^{M} \alpha_k^i \binom{M}{k} s^k (1-s)^{M-k}, \qquad s \in [0,1],$$

where

$$\binom{M}{k} = \frac{M!}{k!(M-k)!}$$

The next step is to generate a target based on the phase of the gait timing variable s where,

$$s = \frac{\theta(q) - \theta^+}{\theta^- - \theta +}$$

The target q is computed using,

$$h_d(\theta(q)) = \begin{bmatrix} b_1 \circ s(q) \\ b_2 \circ s(q) \\ \vdots \\ b_{N-1} \circ s(q) \end{bmatrix} \tag{18}$$

The target $\dot{q}$,

$$\dot{q} = H^{-1} \begin{bmatrix} \dfrac{\partial h_{d,\alpha}}{\partial \theta} \\ 1 \end{bmatrix} \dot{\theta} \tag{19}$$

4 Nonlinear Controller

4.1 Feedback Linearization

4.1.1 Linear Control Using K_p and K_d Gains

A common approach for controlling second-order dynamical systems is the proportional-derivative (PD) controller, which uses proportional gain K_p and derivative gain K_d to regulate the tracking error.
Consider this system

$$\dot{q} = f(x) + g(x)u$$

where q is the system output and u is the control input. Let the tracking error be defined as

$$y = q_d - q$$

where q_d is the desired reference trajectory. The PD control law is given by

$$v = A(K_p y + K_d \dot{y})$$

where $K_p > 0$ and $K_d > 0$ are the proportional and derivative gains, respectively.
Substituting the control law into the system dynamics yields the closed-loop error equation

$$\ddot{y} + K_d \dot{y} + K_p y = 0$$

This represents a standard second-order linear system. The proportional gain K_p increases the response speed and reduces steady-state error, while the derivative gain K_d improves damping and reduces overshoot.
By properly selecting K_p and K_d, the closed-loop system can achieve stable and desirable transient performance such as fast settling time and low oscillation.

5 Results and Discussion

5.1 Optimal Gait Trajectory

This section details the gait trajectory results obtained from the optimization. The results are evaluated in terms of joint kinematics, velocity profiles, and the structure of the resulting limit cycle, providing a comprehensive picture of the emergent locomotion behavior. Figure 2 shown below shows the joint angles over time and displays the personality and characteristic of the gait that has been developed. q_3, or the torso angle, tends to increase and posture upwards while the swing leg moves to the front, and falls immediately back down once the impact has occurred, and the swing leg has switched to the stance position. This process repeats with every gait. Additionally, as expected, the swing phase angle q_2 increases over time, albeit with slight discontinuity; a quick jump in the angle/position at the beginning, and then a more gradual change to the final angular position before foot contact.

5.2 Joint Velocity Profiles

Figure 3 presents the joint velocity profiles in isolation, allowing closer examination of the rate of change of each degree of freedom throughout the step. The velocity of the stance leg angle $\dot{q}_1$ illustrates a smooth, relatively undisturbed and constant velocity over time, which is the desired behavior for the underactuated, cyclic variable. The swing leg joints on the other hand, $\dot{q}_2$ and $\dot{q}_3$, undergo velocity spikes at the beginning of each gait, reflecting the impact reset applied at the beginning of each step.

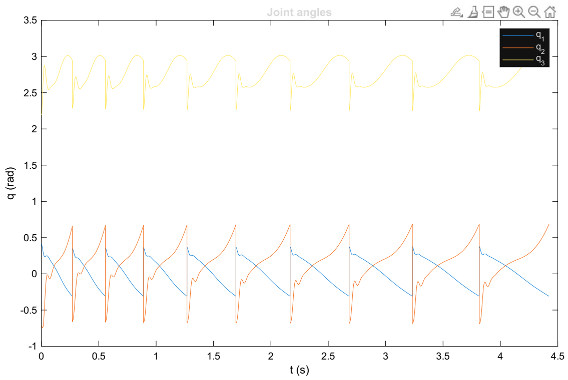

Figure 2: Joint angles over time. The periodic trajectories reflect the Bézier-parameterized virtual constraints enforced during the swing phase.

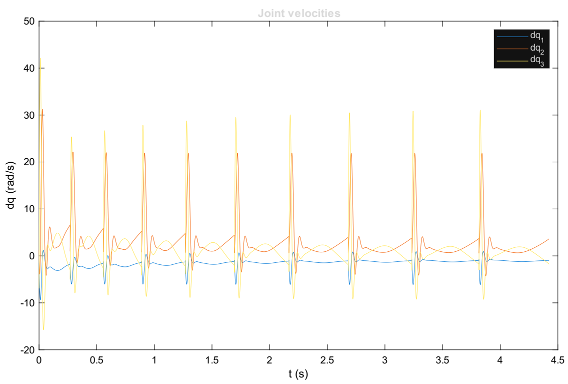

Figure 3: Joint velocity profiles over a single optimized step.

5.3 Phase Portrait — Zero Dynamics

The phase portrait of the zero dynamics model is shown in Figure 4. The closed trajectory in the $(q_1, \dot{q}_1)$ plane confirms the existence of a stable periodic orbit. The simplicity within the shaping of the zero dynamics phasing indicates success of a closed-loop gait, as well as two distinct types of locomotion behaviors within the gait itself that are better contextualized within the full dynamics phase portrait.

5.4 Phase Portrait — Full Dynamics

The phase portrait of the full system dynamics is shown in Figure 5. The closed trajectory in the $(q_1, \dot{q}_1)$ plane here once again confirms the existence of a stable orbit, with the impact map producing a discrete jump in velocity at heel-strike. The shape of the orbit reflects the energy exchange between the stance and swing phases, When checking the animation, it is possible to observe that the swing leg starts

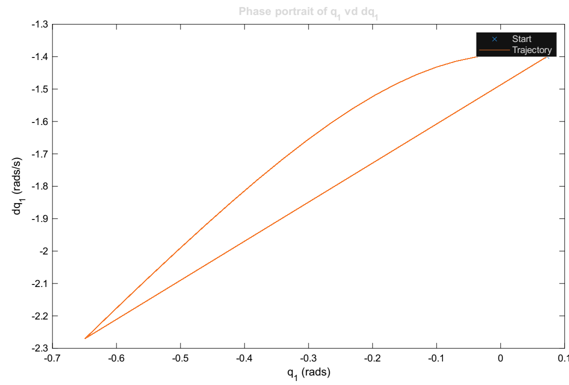

Figure 4: Phase portrait of the zero dynamics model. The closed orbit confirms periodicity of the optimized gait, with the discrete jump corresponding to the heel-strike impact event.

with strength, but then it stops around the middle of the step and then continues along its desired path, causing a sort of limping motion. This would explain why the phase portrait has the p-shape. Even though the motion is cyclical and returns to its desired position every time, the limping motion behavior could be fixed by changing some of the initial guesses of the optimizer, such as q_1 and $\dot{q}_1$, to increase the velocity of the gait, making better steps.

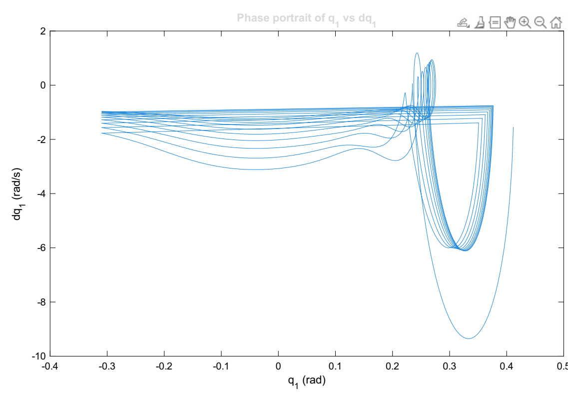

Figure 5: Phase portrait of the full system dynamics. The closed orbit confirms periodicity of the optimized gait, with the discrete jump corresponding to the heel-strike impact event.

6 Future Works

The current three link model works under the assumption that we have rigid, straight walking legs, which limits gait efficiency, and disregards the fact that when walking, the model's 'foot' scuffs during the swing phase. The primary focus of future work will be expanding the current model to a five link bipedal by introducing articulating knee joints, to address current limitations.

The transition to a five link model should drastically improve locomotion, and the addition of knees will allow for active foot clearance during the swing phase, and also introduce a double support phase. However, this model expansion will drastically increase the complexity of the zero dynamics manifold. The state vector q will expand to include 'thigh' and 'calf' relative angles, needing a 5×5 inertia matrix and associated Coriolis effects.

Furthermore, the trajectory generation and optimization will have to be reworked, and the Bézier polynomials that control the virtual constraints will require reworking as well to enforce joint limits to prevent knee hyperextension while minimizing the kinetic energy lost during the impact phase of the limb.

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